Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 9

Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Objective

To implement and analyse the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC) with Ward Linkage.

Dataset

Human Activity Recognition Using Smartphones Dataset (UCI Repository / OpenML)

- 30 volunteers, aged 19–48, performing 6 activities.
- Activities: WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING.
- Full dataset: 10,299 samples, 561 time/frequency-domain features.
- **Working subset: 3,000 samples** (500 per class, stratified sampling).

Preprocessing

1. Load dataset via `fetch_openml('har')` – no missing values.
2. Stratified subsample 3,000 records for tractable runtime.
3. Apply `StandardScaler` (zero mean, unit variance).
4. Dimensionality reduction with PCA: 50 components (87.5% variance) for clustering; 2 components for 2D scatter visualisation.

K-Means – Elbow Method Results

Table 1: K-Means Elbow Method Results (3,000 samples, PCA-50)

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	752,291.5	0.4360
3	640,207.5	0.3575
4	601,425.9	0.2132
5	563,592.2	0.1750
6	541,905.4	0.1485
7	519,818.5	0.1366
8	505,576.6	0.1242

Chosen k: 6 (matches the 6 true activity classes). Although k=2 has the highest silhouette score, k=6 aligns with domain knowledge and the elbow shows diminishing WCSS returns beyond k=6.

Evaluation Metrics

Table 2: Clustering Performance Comparison (3,000 samples)

Algorithm	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means ($k = 6$)	**0.1485**	2.1386	**1027.6**	**0.2822**	**0.4538**
DBSCAN ($\epsilon = 16$)	N/A*	N/A	N/A	0.00	0.00
HAC (Ward, $k = 6$)	0.1247	2.2977	958.9	0.2959	0.4624

*DBSCAN found 2 clusters + 179 noise points; low quality metric

DBSCAN Parameter Tuning

- k-NN distance analysis (k=15) yielded: median = 11.1, p90 = 17.0, p95 = 20.2 in PCA-50 space.
- Best result: $\epsilon = 16$, $minPts = 10 \rightarrow 2$ clusters, 179 noise points (6% noise).
- DBSCAN is not well-suited to this dataset because HAR features occupy a roughly uniform density in PCA space – DBSCAN cannot distinguish the 6 activities as separate density regions.

Graphs and Visualisation

All figures saved to `plots/`:

- **kmeans_elbow_silhouette.png** – Elbow and Silhouette vs. k curves.
- **true_labels_pca.png** – Ground-truth activities in PCA-2D.

- **kmeans_clusters.png** – K-Means ($k = 6$) clusters in PCA-2D.
- **dbscan_clusters.png** – DBSCAN ($\epsilon = 16$) clusters in PCA-2D.
- **hac_clusters.png** – HAC (Ward, $k = 6$) clusters in PCA-2D.
- **dendrogram.png** – Ward linkage dendrogram (300-sample subset, truncated).
- **metrics_comparison.png** – Bar chart comparing metrics across algorithms.

Observation Questions

- **Which algorithm produced the most meaningful clusters?** K-Means and HAC both produced comparable results (ARI $\approx$ 0.28–0.30, NMI $\approx$ 0.45). HAC marginally edged K-Means on ARI/NMI, while K-Means had a slightly higher Silhouette and Calinski-Harabasz score.
- **How sensitive was K-Means to the choice of k?** The Silhouette score steadily decreases from $k = 2$ to $k = 8$, suggesting the data does not have strongly separated high-dimensional clusters. The elbow in WCSS is mild; k=6 is chosen by domain knowledge.
- **Did DBSCAN detect noise or small clusters effectively?** DBSCAN could not partition the 6 activities – it collapsed most data into 1–2 large clusters with significant noise. HAR features form a continuous manifold in PCA-50 space without clear density gaps.
- **How does linkage choice affect HAC?** Ward linkage minimises within-cluster variance (similar to K-Means objective) and produced the best HAC results. Single linkage would suffer from "chaining"; complete linkage would be more compact but less balanced.
- **Which internal metric best matched visual intuition?** Calinski-Harabasz index confirmed K-Means > HAC. The Silhouette score's low absolute values ($\approx$ 0.15) correctly reflected the visual observation that PCA-2D clusters overlap – particularly the locomotion activities (WALKING, WALKING_UP, WALKING_DN) which share similar sensor signatures.

Conclusion

K-Means ($k = 6$) and HAC (Ward, $k = 6$) both achieved moderate agreement with ground-truth activity labels (NMI $\approx$ 0.45), demonstrating that unsupervised methods can capture meaningful structure in HAR data without labels. DBSCAN was unsuitable for this high-dimensional, uniformly-dense dataset. Dimensionality reduction with PCA to 50 components preserved 87.5% of variance and enabled tractable clustering.

References

- UCI HAR Dataset
- Scikit-learn Clustering Documentation